BIG DATA AT THE EDGE • PROJECT PRESENTATION

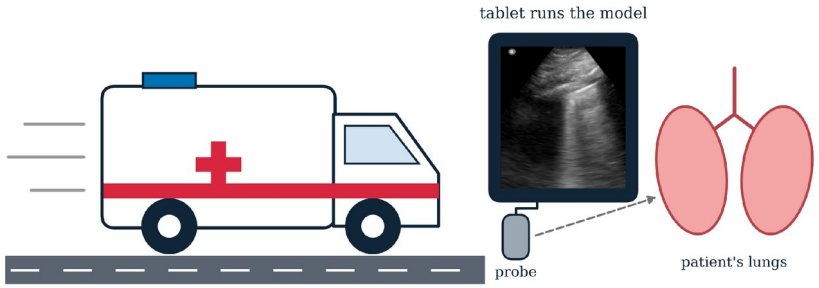
Adaptive inference for lung ultrasound at the edge

Group 7 | Charalampos Giannakis & Salomé Poulain

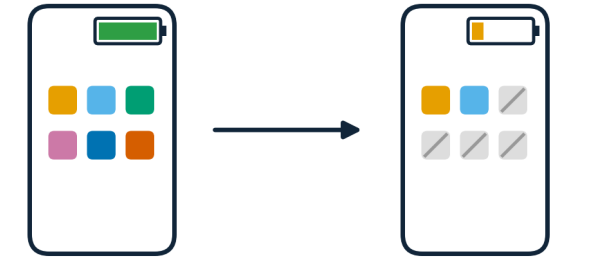
Question: can one model adjust computation to a changing resource budget without giving up useful predictive quality?

VARIABLE BUDGET → SHARED-WEIGHT MODES → MEASURED TRADE-OFF

Problem Scenario



Phone low-power mode: things off, quality down



full battery
everything runs

low-power mode
background apps off

SCENARIO

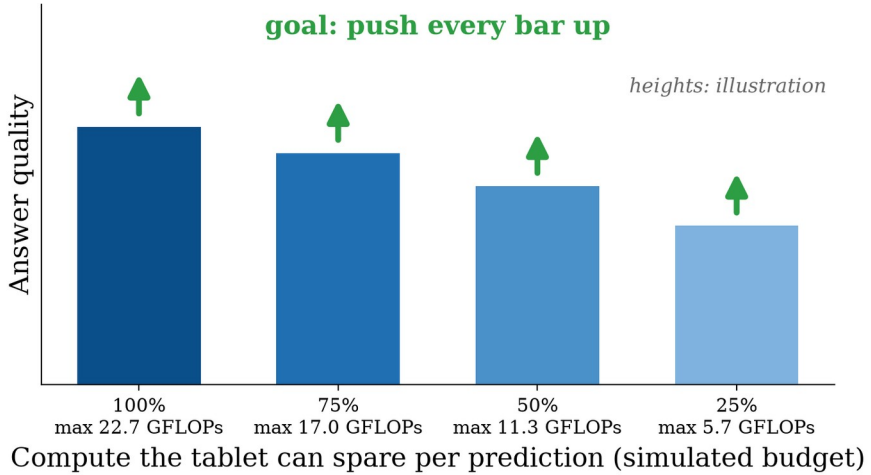
Tablet flags abnormal lung zones

Spare compute changes with heat, the ultrasound app, battery

Big fixed model: only fits at 100%

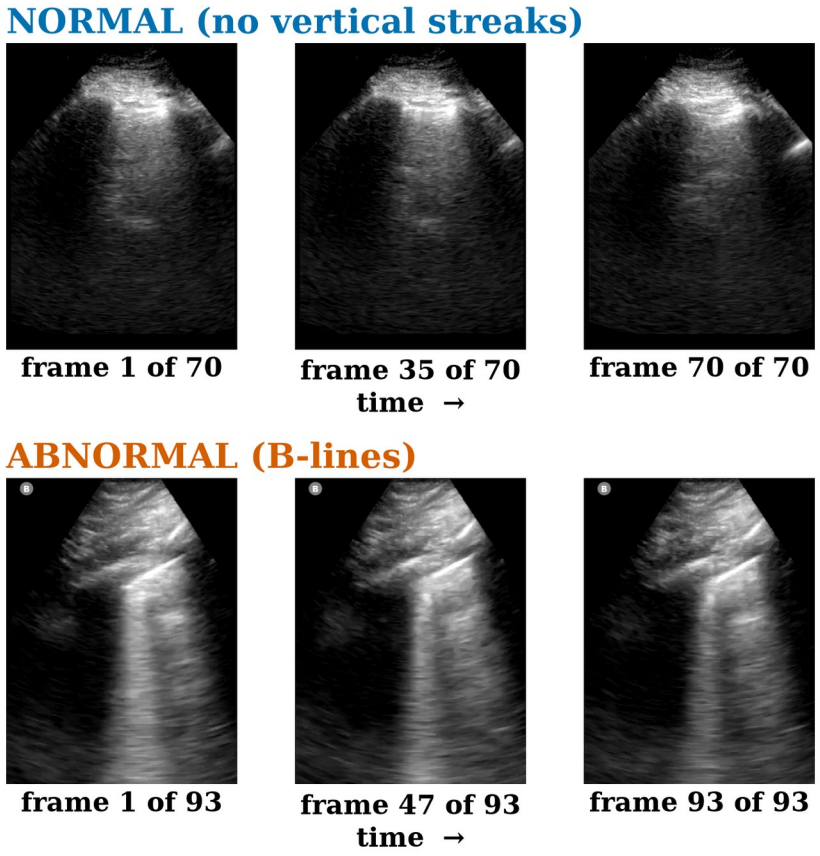
Small fixed model: never uses the extra

GOAL



Goal: choose a useful quality–cost mode within each budget.

Dataset and Task



DATA

The data

Public Lung Data was rare.

OpenPOCUS

Researchers at Stanford published it in 2026 for training NN

~50% potential COVID-19, 50% control

226 exams ~ 12 zones

1 video = 1 label: *normal*, *3x non-normal findings* or *can't agree*

binary: normal or abnormal

1,762 videos

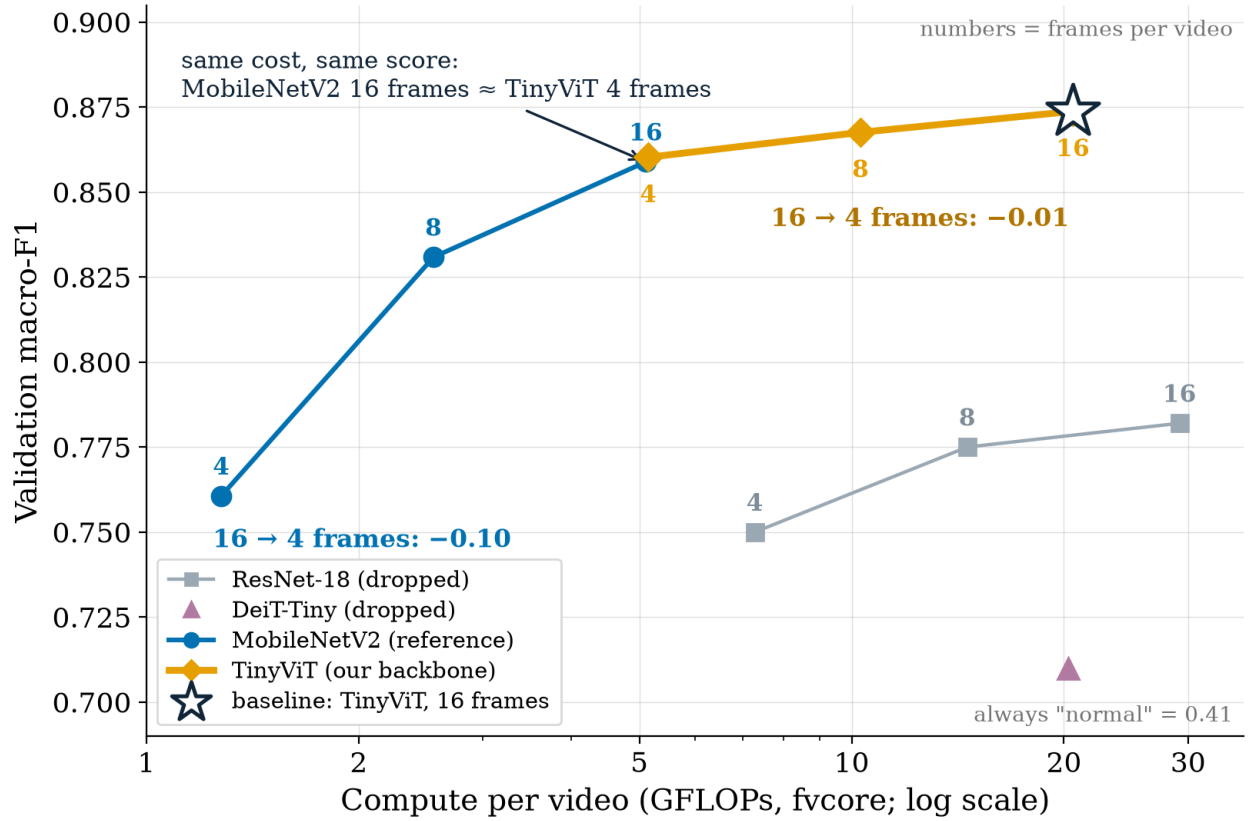
70% normal (macro-F1 needed)

stratified group split: 70/15/15

Max 16 spread out frames

POCUS stands for point-of-care ultrasound: a doctor scans the patient at the bedside with a handheld or portable probe (Kumar et al., POCUS Journal, 2026)

Baseline and Backbone



SETUP

Four models tested

Beat: always normal 0.41 M-F1

Round 1: 16 frames

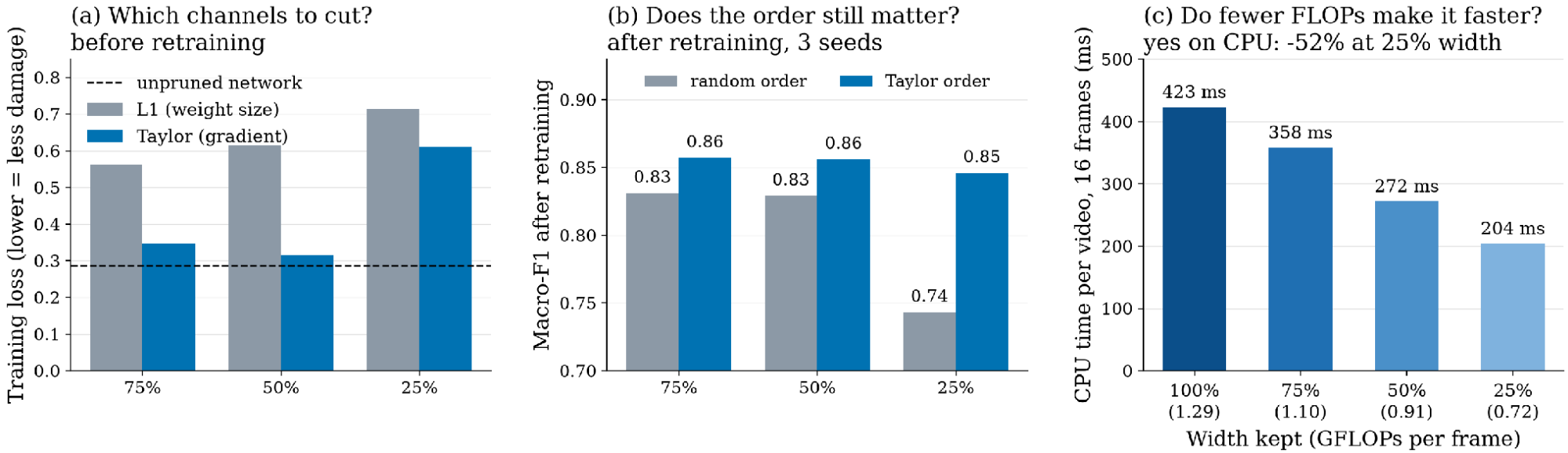
Round 2: test 8, 4 frames

TinyViT vs. MobileNetV2:

- frame knob: -0.01 loss vs. -0.10
- same compute cost, same score
- more GFLOP room to scale down

Backbone: TinyViT
Baseline: TinyViT 16 frames
Reference: MobileNetV2

Structured Pruning



Tried first: 8-bit: 3x smaller file, same FLOPs

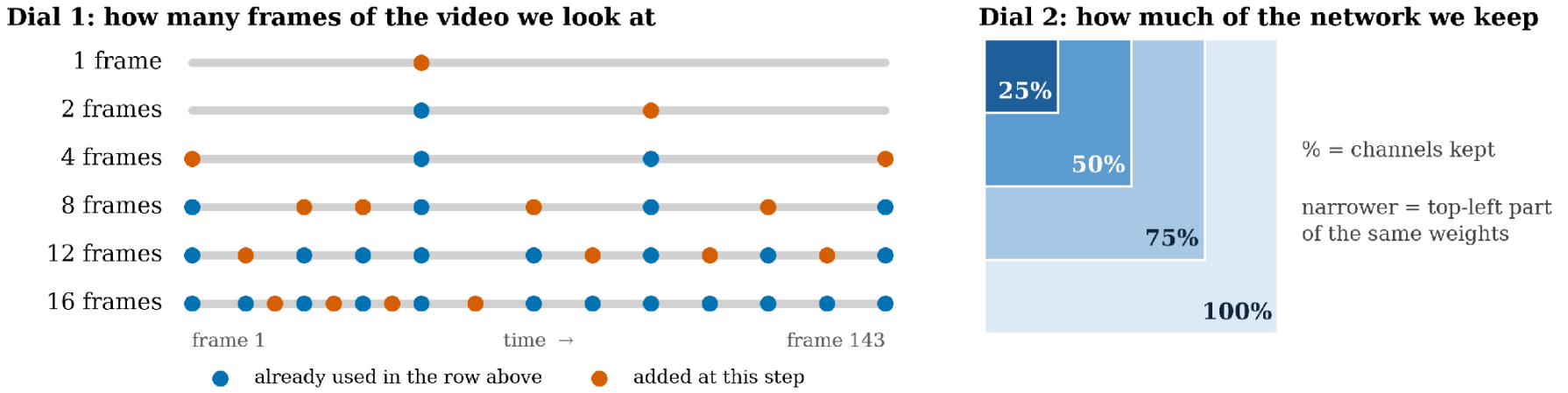
Then: cut whole inner channels (59% of compute)

Nested: sorted by Taylor, keep 25–100% in one file

Retrained: knowledge distillation, frozen teacher

Second knob: 25% width close on average, 2× faster at 16 frames

Adaptive Mechanism



1. Before: train one network for all 24

Every training step: same videos at 3 widths (full, 25%, and 50 or 75%) sandwich rule

Each width: answers from first 1, 2, 4, 8, 16 frames

Distillation with Baseline 16 teacher per frame setting

Checked in torch: all widths share the weights

2. In use: pick 1 of 24 per video

Dial 1: 1, 2, 4, 8, 12 or 16 frames

Less budget: drop the orange frames

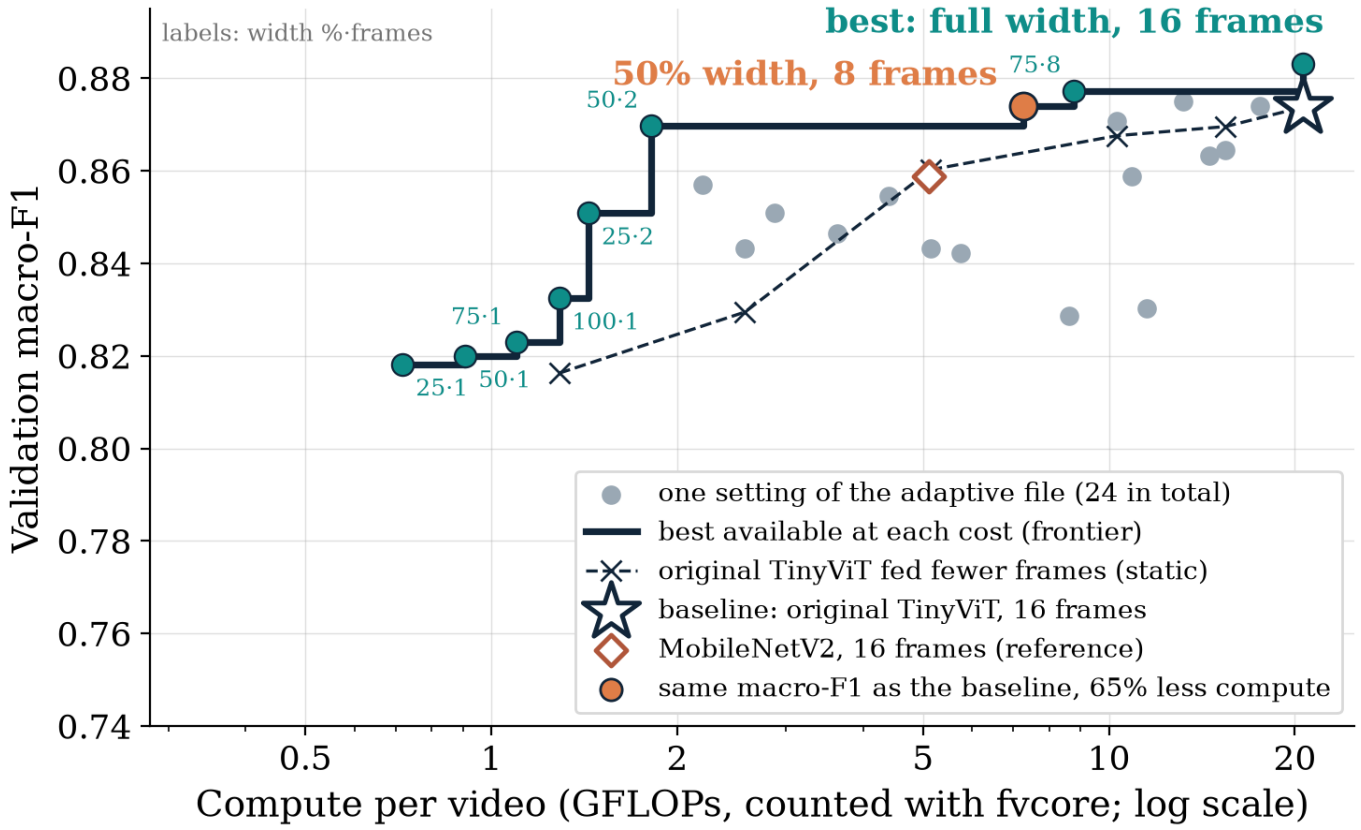
Dial 2: 25, 50, 75 or 100% of channels

Less budget: drop the least important channels

Switching width: under 1 ms

24 configurations in one 20.4 MB file

Pareto Frontier: macro-F1 vs. FLOPs



EVIDENCE

Result

Frontier above Baseline and MobileNetV2

Orange: baseline score, ⅓ of the compute

Full width above star: 2 videos, extra training

whole range, from 0.7 to 20.6 GFLOPs

Controller Policy

External budget b → eligibility screen → deterministic selection or deferral.

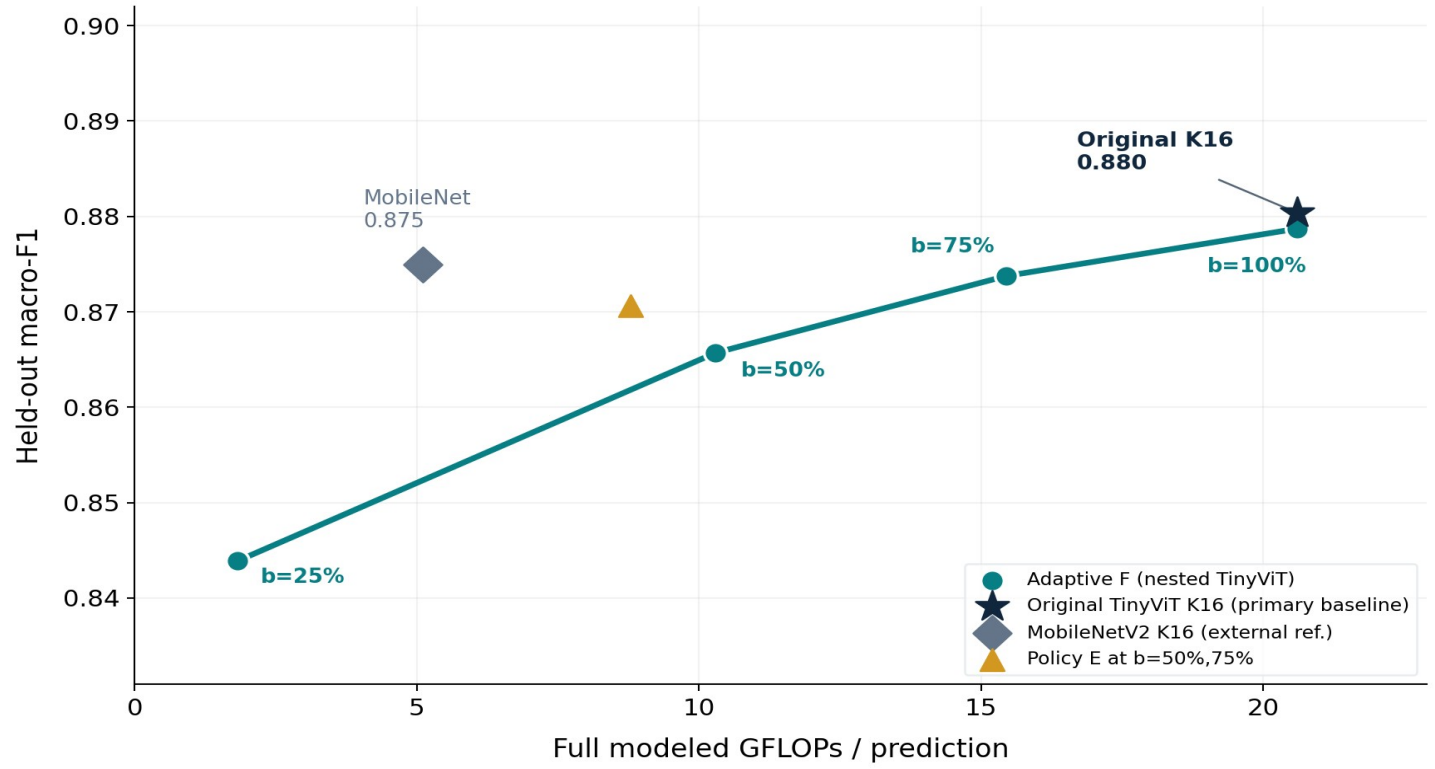
Budget b → quality filter → exact FLOP cap → max abnormal recall → lower cost

B = 0.00	B = 0.25	B = 0.50	B = 0.75	B = 1.00
DEFER	w0.5 · K2	w1.0 · K8	w1.0 · K12	w1.0 · K16
0 G	1.82 G	10.30 G	15.45 G	20.60 G

A simulated FLOP budget, not a measured battery, latency, or thermal constraint.

Test Set Results

265 unseen test units · all shown positive-budget points have 100% coverage



ONE CHECKPOINT

91% less

modeled FLOPs at b=25%
F1 0.844 vs original 0.880

AT FULL COMPUTE

0.879 vs 0.880

Nested vs original TinyViT
Same 20.60 GFLOPs; not
a formal equivalence result.

Original TinyViT K16 is the main architectural baseline; MobileNetV2 is a separate external reference.

Mixed Budgets

Each frozen test unit gets one simulated budget state · 53 units per state · 0% always defers

Predictions completed under the same resource schedule



Comparison is coverage, not evidence that F has higher accuracy on the same units.

CONTROLLER REALITY CHECK

E and F both cover 80%

Under this mixed schedule:

E · 78.6% abnormal recall

8.00 GFLOPs / request

F · 76.8% abnormal recall

9.63 GFLOPs / request

E used less compute with higher observed recall.
F was not retuned on test.

MobileNetV2 (A) also reaches 80% coverage and uses fewer FLOPs than F.

CPU Latency

Validation benchmark · JPEG preprocessing + model inference · one laptop CPU, four threads

Median latency per sequence (ms)



200 validation requests · 50 warmups · 5 repeats · AMD Ryzen AI 9 HX 370

No measured energy or battery performance; this is not an ambulance device.

Conclusion and Reflection

The flexible model family works; selecting the best resource policy is a separate problem.

BUILT

Shared-weight TinyViT with 24 configurations on a budget routing

OBSERVED

Requests completed:
Original TinyVit → 20%
Adaptive → 80%

LEARNED

Recall-priority F did not beat macro-F1 policy E on held-out recall

MobileNetV2 also reaches 80% coverage · OpenPOCUS proxy ≠ clinical validation · FLOPs ≠ energy

Questions

&

Answers

BACKUP • Why did policy E beat recall-priority F?

The controller optimizes VALIDATION recall; held-out rankings need not match.

At b=.50: F recall .8025 at 10.30 G | E .8148 at 8.78 G
At b=.75: F recall .8025 at 15.45 G | E .8148 at 8.78 G
Mixed: F recall .7679 at 9.63 G | E .7857 at 8.00 G

At b=.25 and b=1.00, F and E make identical predictions.
F has slightly higher macro-F1 at b=.75; no all-metric winner.

Frozen choices; no test-set retuning. 95% paired intervals are unadjusted.

BACKUP • What the held-out test establishes

One authorized, frozen evaluation • not new clinical evidence

265 held-out recording units (81 abnormal, 184 normal)
32 known dependency groups • 5,000 paired bootstrap draws
38 evaluation cells • 24 prespecified contrasts

Static TinyViT K16: 20% mixed-budget coverage
Adaptive TinyViT F: 80% coverage
MobileNetV2 A: also 80% coverage, at lower modeled cost

Quality only on served units; paired contrasts use shared served units.

Known-group protection ≠ known distinct patients; OpenPOCUS remains a technical proxy.

BACKUP • Why 16 spread frames?

16 was set before training • afterwards we tested whether it holds up

what we tried	result	source
All frames (median ~143 per video) vs 16	no better: macro-F1 0.870 vs 0.874, at ~184 vs 20.6 GFLOPs	F004
Fewer frames: 1, 2, 4, 8, 12, 16	8 ≈ 16, 4 slightly lower, 1 and 2 clearly lower	F003
Spread vs consecutive (4 in a row vs 4 spread out)	spread better at 4 ($p = 0.04$); smaller at 8, none at 16	F003
Coverage of the spread (first 25, 50, 75 or 100% of clip)	no measurable difference	F003

All on validation (254 units) • TinyViT trained at 16 frames, tested at fewer without retraining